

1

From the given data,

$$\frac{k}{m} = \omega_n^2 = (2\pi f_n)^2 = (8\pi)^2 = 64\pi^2 \text{ --- (a)}$$

$$\frac{k}{m + \Delta m} = (\omega_n')^2 = (2\pi f_n')^2 = (6\pi)^2 = 36\pi^2 \text{ --- (b)}$$

Dividing eqn (a) by eqn (b) gives

$$1 + \frac{\Delta m}{m} = \frac{16}{9}$$

$$m = \frac{9}{7} \Delta m = \frac{9}{7} * \frac{5}{9} = \frac{6.43}{9} \text{ lbs/ft} \text{ --- (c)}$$

From eqn (a)

$$k = 64\pi^2 m = 64\pi^2 * \frac{6.43}{9} = \underline{10.52 \text{ lbs/in}}$$

2

At $\omega = \omega_n$; [from eqn from Text book (3.2.15)]

$$u_0 = (u_{st})_0 \frac{1}{2\xi} = 2 \text{ --- (a)}$$

At $\omega = 0.1 \omega_n$; [from eqn from Text book (3.2.13)]

$$u_0 \approx (u_{st})_0 = 0.2$$

Substituting $(u_{st})_0 = 0.2$ in eqn (a) gives,

$$\underline{\xi = 0.05}$$

3] Given,

$$W = 1200 \text{ lbs}; \quad E = 30 \times 10^6 \text{ psi}$$

$$I = 10 \text{ in}^4; \quad L = 8 \text{ ft}; \quad \xi = 1\%$$

$$P_0 = 60 \text{ lbs}; \quad \omega = \left(\frac{300}{60}\right) 2\pi = 10\pi \text{ rad/s}.$$

Stiffness of Two beams:

$$K = 2 \times \left(\frac{48EI}{L^3} \right) = 32,552 \text{ lbs/in.}$$

Natural frequency:

$$\begin{aligned} \omega_n &= \sqrt{\frac{K}{m}} = \sqrt{\frac{K}{W/g}} = \sqrt{\frac{32,552}{1200/386}} \\ &= \underline{102.3 \text{ rad/sec.}} \end{aligned}$$

Steady state response

$$R_d = \frac{1}{\sqrt{\left(1 - (\omega/\omega_n)^2\right)^2 + \left[2\xi\omega/\omega_n\right]^2}}$$

$$\text{where, } \omega/\omega_n = 10\pi/102.3 = \underline{0.3071}.$$

$$Q_d = \frac{1}{\sqrt{(1-0.0943)^2 + (2 \cdot 0.01 \cdot 0.3071)^2}} = 1.104$$

Displacement:

$$u_0 = (u_{st})_0 \cdot R_d = \frac{P_0}{K} \cdot R_d = \frac{60}{32,552} \cdot 1.104$$

$$= 2.035 \cdot 10^{-3} \text{ in}$$

Acceleration amplitude:

$$\ddot{u}_0 = \omega^r u_0 = (10\pi)^r \cdot 2.035 \cdot 10^{-3}$$

$$= 2.009 \text{ in/sec}^r = 0.0052 g$$

4] (a) Natural frequency

The frequency response curve peaks at:

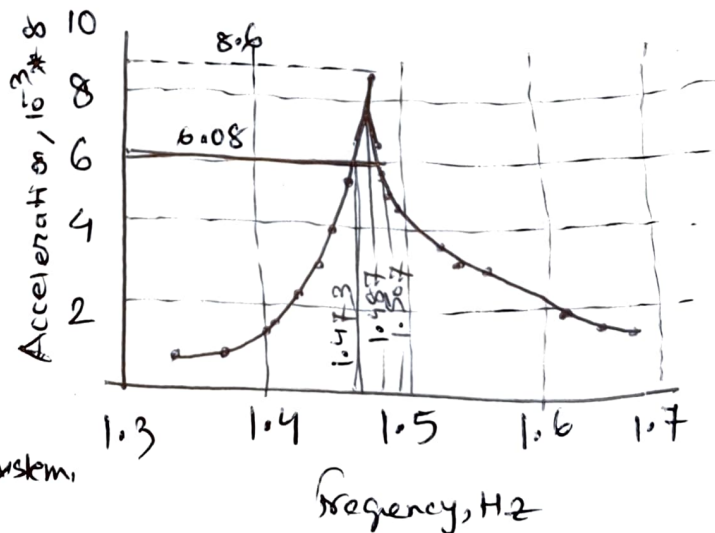
$$f_n = 1.487 \text{ Hz}$$

Assuming small damping, this value is the natural frequency of system.

(b) Damping Ratio

The acceleration at the peak is $r_{\text{peak}} = 8.6 \cdot 10^{-3} g$.

Draw a horizontal line at $r_{\text{peak}} / \sqrt{2} = 6.08 \cdot 10^{-3} g$ to obtain f_a & f_b in Hz



Then,

$$\xi = \frac{f_b - f_a}{2f_n} = \frac{1.507 - 1.473}{2 \times 1.487} = 0.0114' \\ = 1.14\%$$